8.8 Salt Hydrolysis

When an acid reacts with a base, a salt and water are formed and the reaction is called neutralization. Salts completely dissociate in aqueous solutions to give their constituent ions. The ions so produced are hydrated in water. In certain cases, the cation, anion or both react with water and the reaction is called salt hydrolysis. Hence, salt hydrolysis is the reverse of neutralization reaction.

8.8.1 Salts of strong acid and a strong base

Let us consider the reaction between NaOH and nitric acid to give sodium nitrate and water.

$$NaOH(aq)+HNO_3(aq) \rightarrow NaNO_3(aq)+H_2O(l)$$

The salt $NaNO_3$ completely dissociates in water to produce Na^+ and NO_3^- ions.

$$NaNO_3(aq) \rightarrow Na^+(aq)+NO_3^-(aq)$$

Water dissociates to a small extent as

$$H_2O(l) \rightleftharpoons H^+(aq)+OH^-(aq)$$

Since $[H^+]=[OH^-]$, water is neutral

NO_3^- ion is the conjugate base of the strong acid HNO_3 and hence it has no tendency to react with H^+.

Similarly, Na^+ is the conjugate acid of the strong base NaOH and it has no tendency to react with OH^-.

It means that there is no hydrolysis. In such cases $[H^+]=[OH^-]$ pH is maintained and, therefore, the solution is neutral.

8.8.2 Hydrolysis of Salt of strong base and weak acid (Anionic Hydrolysis).

Let us consider the reactions between sodium hydroxide and acetic acid to give sodium acetate and water.

$$NaOH\ (aq) + CH_3COOH(aq) \rightleftharpoons CH_3COONa(aq)+H_2O(l)$$

In aqueous solution, CH_3COONa is completely dissociated as below

$$CH_3COONa\ (aq) \longrightarrow CH_3COO^-(aq)+Na^+(aq)$$

CH_3COO^- is a conjugate base of the weak acid CH_3COOH and it has a tendency to react with H^+ from water to produce unionised acid .

There is no such tendency for Na^+ to react with OH^-.

$CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH\ (aq) + OH^-(aq)$ and therefore $[OH^-]>[H^+]$, in such cases, the solution is basic due to hydrolysis and the pH is greater than 7.

Let us find a relation between the equilibrium constant for the hydrolysis reaction (hydrolysis constant) and the dissociation constant of the acid.

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